

Pip Tutor — Turn Latency

Record-end → robot starts speaking the feedback. ESP32-S3, home Wi-Fi (Israel), functions in `europa-west10` .

ORIGINAL (MP3 BASELINE)

14.8–19s

NOW (PHASE 2)

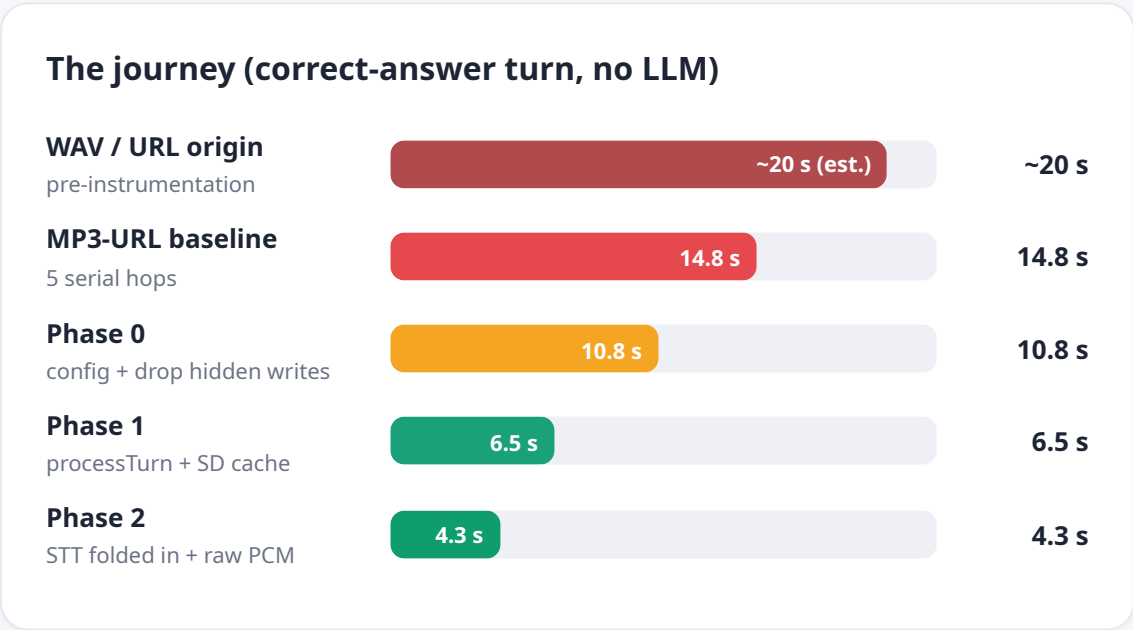
4.0–5.8s

IMPROVEMENT

~3.5× faster

CUT PER TURN

10–13s



What each iteration changed

ITERATION	KEY LEVER	CORRECT	WRONG (GEMINI)	HOPS
WAV / URL origin	Full uncompressed WAV downloaded before playback	~20 st	—	5

ITERATION	KEY LEVER	CORRECT	WRONG (GEMINI)	HOPS
MP3-URL baseline	MP3 over URL; write→trigger→poll→download	14.8 s	17–19 s	5
Phase 0	minInstances , preferRest , thinkingBudget:0 , keep-alive poll, drop 2 cosmetic writes	10.8 s	12.5 s	5*
Phase 1	One synchronous processTurn — grade + gated Gemini + TTS, MP3 in the response body; SD- cached prompts	6.3–6.6 s	7–8.3 s	2
Phase 2	Upload raw PCM to processTurn (STT server-side); keep-alive heartbeat	4.0–4.3 s	5.8 s	1

† Not instrumented (timers added at the MP3 baseline). * Still 5 hops, but warm — the up-to-+12 s cold-start cliff is gone. Structural win: **5 → 2 → 1** network hops; inline-audio decode dropped from a 2.6–3.8 s URL download to **2–4 ms**.

Phase 2, measured live on hardware

TURN	HEARD	PATH	PROCESSTURNAUDIO	DECODE	RECORD→SOUND
1	"sister" ✓	grade only	4341 ms	2 ms	4343 ms
2	"milk" ✓ (streak)	grade only	4066 ms	4 ms	4073 ms
3	"no idea" ✗	grade + Gemini hint	5841 ms	2 ms	5843 ms

The remaining ~4 s is essentially the single STT + compute + TTS round trip Israel↔Berlin — close to the practical floor without changing region. Turn 3's extra ~1.5 s is the (correctly gated) Gemini hint call.

Generated 2026-06-29 · see [Documentation/LATENCY.md](#) for the hop-by-hop analysis.